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01 Working on: "Reading Research paper"  
Reliability Concerns in R TSV based 3D  
Integration: Fabrication to packaging.

01 what is Fabrication:

definition: building things out of raw material

(making IC on a silicon wafer <sup>large plate</sup>)

02 what is packaging: Enclosing or re-connecting the chip. so it can be used in a electronic system.

(Die's or chips from the wafer) and

(its protects or allow that chip to connect the circuit)

### Abstract

01 heterogeneous integration techniques; combining diff. types of electronic components into one single unit.

What is the problem: TSV faults must be screened promptly during the manufacturing process. (because at that time period the mech. stress and chip packing are considered as being major cause 3D TSV IC failures).

What the discussed:

01 various 3D TSV IC defects from fabrication to packaging.

02 Also discussed several defect mechanisms & fault structures of packaging, and reliability issues.

What paper tells:

01 different types of defects from fabrication to packaging.

01 Types of defects.

01 void (fabrication)

02 delamination

03 open defect (break in an elec. path) <sup>bump cut. for suppose the path got cut.</sup>

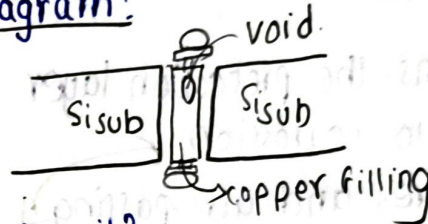
04 Anhole defect (fabrication)

05 short defect, (metal spreading out via)

1.1 void defect (fabrication)

definition: Tiny empty gap inside through silicon via.

Diagram:



what where mechanism

what is it?

1 TSV is supposed to be filled with metal (usually copper)

2 during fabrication the metal doesn't fill properly leaves hallow space.

Mechanism:

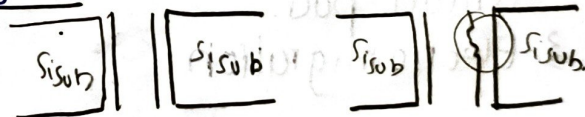
1 poor metal filling

2 shrinkage while cooling.

1.2 Delamination (cracks) (packaging)

definition: layers inside the chip starts to break due to thermal expansion.

Diagram:



what is it:

1 The cracks are formed by expansion  
2 It can cause [the signals can't pass properly]

Mechanism:

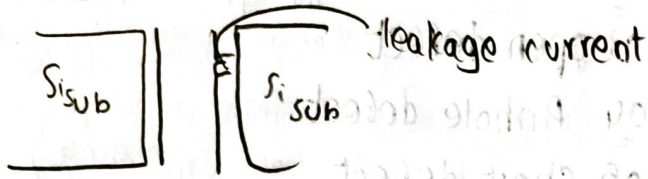
1 Thermal stress, Mechanical stress

2 Moisture & contamination



### 1.3 Pin hole effect (short defect)

definition: Pin hole formation results in a leakage btw TSV and the Si<sub>sub</sub>.



where it shows up:

Near insulating layer around TSV

why it's a problem:

1. causes short circuits in electronics.

2. weakens the protection layer.

3. leads to corrosion

4. Moisture and air passing

Mechanism:

1. dust or particle during the manufacturing.

2. Improper coatings

### 1.4 Co-pumping:

Electric failures can occur as Cu diffuses to the silicon sub. through insulating. (pin hole)

### 02 Impact of packaging.

1. Disconnection failure

2. Misalignment of TSV, Bump and contact pad.

3. Electromigration.

### 2.5 Disconnection failure.

definition: disconnection failures are caused by impurities btw the TSV & bumps & adjacent bumps which breaks the e<sup>-</sup> flow "I".

02 This can ↑ RC paracitics

03 if the e<sup>-</sup> flow stopped and signals stopped [open circuit] & charge will start to build up. (that can cause voltage spikes.

↳ Instead of staying smooths it jumps with 1V

### 2.3 Misalignment:

short bumps with medium TSV.

low contact A ↓ R ↑ this also causes voltage spikes

### 2.4 Electro migration:

01 what is it?

a lot of I through thin metal wires

→ e<sup>-</sup> pushes and moves from original position.

→ wire gets damaged overtime

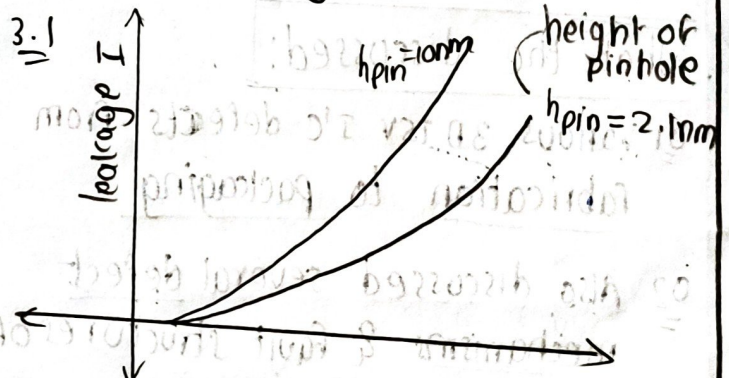
02 leads to

1. void formation

2. unwanted short circuits.

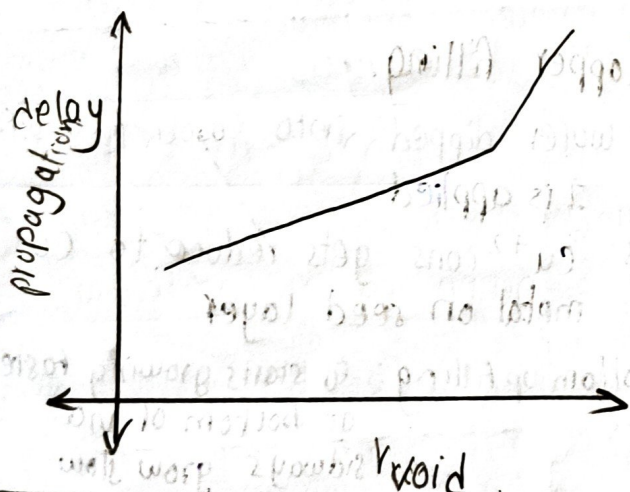
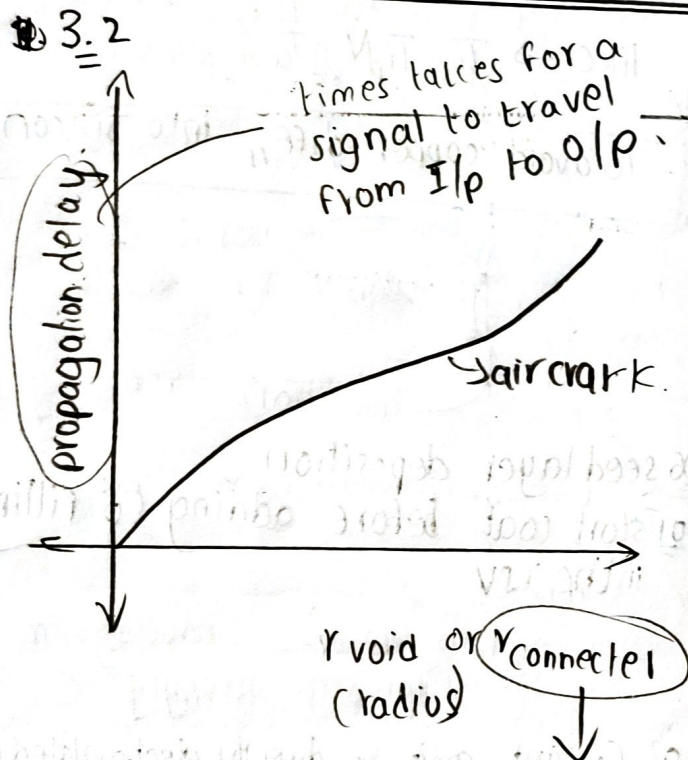
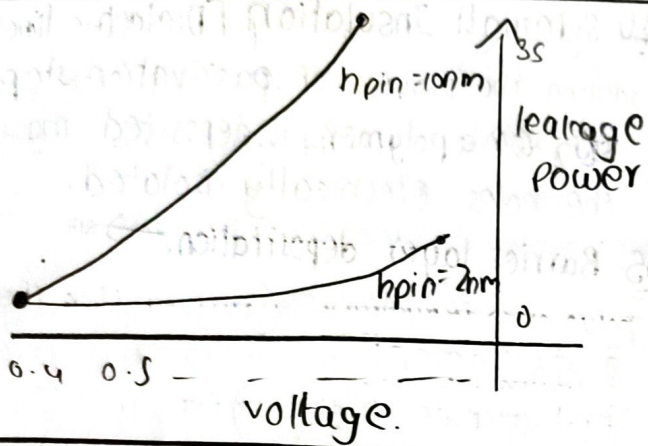
3. R ↑ parasitic.

### 03 Defect Analysis.



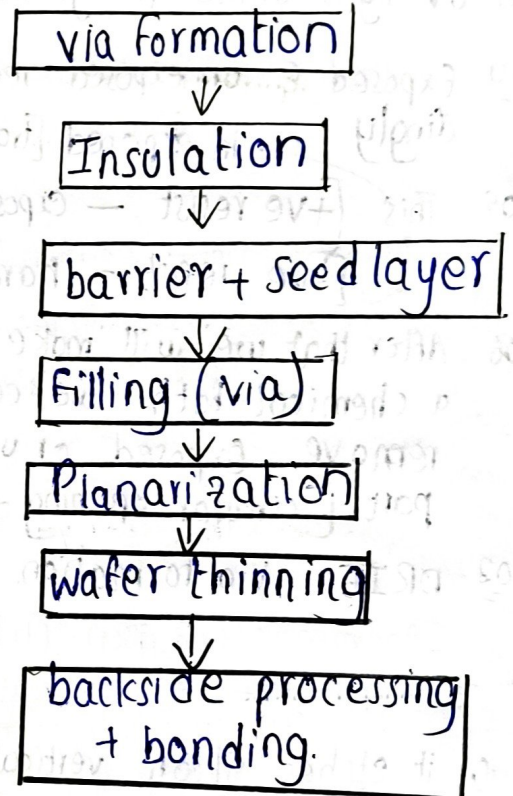
h<sub>pin</sub> ↑ leakage I ↑





- 01 TSV fabrication process
- 02 more papers [pin hole effect]
- 02 Working on: "TSV fabrication process"

## Fabrication Flow

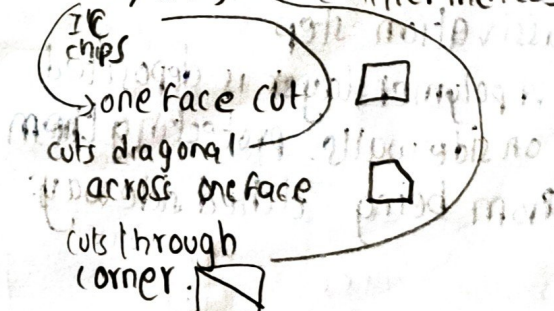


Via formation Process: silicon circular plate.

01 wafer preparation (cleaning, its orientation and thickness is checked)

orientation: how the crystal lattice of the silicon is aligned when wafer is sliced.

→ 100, 110, 111 (Miller indices)





## Lithography (via locations in stacking)

- 01 silicon wafer coated with a light sensitive material [photoresist]
- 02 A photomask gets aligned on the top with that mask having transparent and opaque regions.
- 03 UV light shines through mask.
- 04 Exposed & unexposed react accordingly in exposed [transparent]
- 05 This [+ve resist - exposed dissolve] [-ve resist - harden]

06 After that we will make the wafer a chemical bath we can either remove exposed or unexposed part [circular opening - viapattern]

## 02 DRIE - via formation.

[Deep reactive Ion Etching]

[using plasma instead of mech., drill]

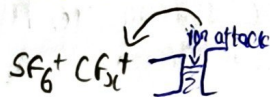
01 it etches silicon vertically downward with very high aspect ratio

how process  
plasma

depth  $\rightarrow 10,000 \mu m$   
width  $\rightarrow$  few  $\mu m$   
ratio: 50:1

### 1.1 Etch step

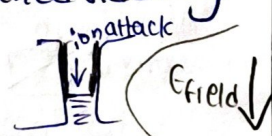
- $\rightarrow$  plasma ions attack silicon.
- $\rightarrow$  removes the material straight down.



### 1.2 Passivation step

- $\rightarrow$  A polymer layer is deposited on side walls, protecting them from being etched sideways.

(Si-Si ions either knocked out [sputtering] or reactive with  $SiF_4$  & gets evaporated)



## 04 Sidewall Insulation [Dielectric liner]

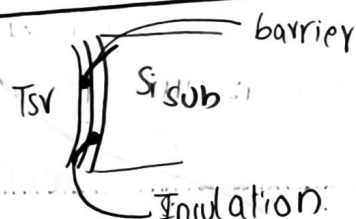
$\hookrightarrow$  during the process of passivation step ~~SiO2~~ a polymer is deposited, making the holes electrically isolated.

05 Barrier layer deposition.  $\rightarrow$  anti polymer  $\rightarrow$  temporary (just to stop the shuttling silicon atoms to stick sides or evaporate Si atoms)

like  $\rightarrow$  Ti, TiN, Ta Tantalum

Titanium Titanium nitride

To avoid copper diff. into Silicon.



06 seed layer deposition.

01 start coat before adding Cu filling in the TSV

$\rightarrow$  A very thin, continuous conducting film. so the Cu can stick strongly.

02 Cu cant. can't be directly electroplated on insulation or barrier

## 03 Copper filling.

01 wafer dipped into  $CuSO_4$  (Cu sulfate)

02 It is applied

03  $Cu^{+2}$  ions gets reduce to Cu metal on seed layer

Bottom up filling: Cu starts growing faster at bottom of via sideways grow slow avoids trapping air voids

## 08 Annealing Cu

(we heat the wafer in a controlled way to settle the copper and fix its internal stress)



## 09 Chemical Mechanical polishing.

- 01 cleaning the surface of wafer
  - 1.1 Extra Cu removed
  - 1.2 Surface flattened by removing Cu Extra.
  - 1.3 Exposes TSV tips.

## 10 wafer thinning.

- 01 get's thinned from backside.

## 11 Backside processing (final preparation)

- 01 Etching from backside of dielectric

- 02 opening TSV tips.

- 03 adding RDL

## 12 Bonding / 3D Integration:

- 01 chips are stacked.
  - wafer to wafer
  - Die to wafer. etc.

after thinning the barrier, Ins. layers can get smugged and forms a circular layer so again we will [drill].

th parameter values with those 3D electromagnetic simulator simulations.

- 03 Impacts of defects on transmission performance of TSV signals.

- ↳ by varying
  - 1 dimension of leakage factor.
  - 2 position of leakage factor.

- 04 Also the coexistence impact of void & leakage on TSV signal also investigated.

## Parameters

- 01 via first, middle, last. 1st slide.
- 02 flow chart 2nd slide
- 03 defects.
- 04 pin hole defect why?

pp 7

nsn model.

## 03 Working on:

"Modelling and simulation of TSV considering void and leakage defects"

Parameter values:

- 10 Establish equivalent circuit model.

leakage defect TSV + void defect TSV + defect free TSV

- 02 This circuit is simulated & its accuracy is confirmed by

Title

Report

Idea:

1 Pinhole defect Exp. & why focusing on that defect

Methodology

Base paper - novel TSV